Out[17]:

Figure 2: Plotted station map

Step 5: Plotting the stations

Suppose you don't want a heat map, but rather would like to plot all the stations. This can be easily done in Folium.

We again use the *Map* class to create a *Map* object. As before, we will provide a value for the parameter *location*. Note that the default value for the parameter *tiles* is *OpenStreetMap*. In the *Map* I created, I used the value *Stamen-Terrain*. You can experiment with various tiles, as given below:

```
"OpenStreetMap"
"Mapbox Bright" (Limited levels of zoom for free tiles)
"Mapbox Control Room" (Limited levels of zoom for free tiles)
"Stamen" (Terrain, Toner, and Watercolor)
"Cloudmade" (Must pass API key)
"Mapbox" (Must pass API key)
"CartoDB" (positron and dark_matter)
```

Having created a *Map* object, you can use the *Marker* class to add points to the map. We will add the markers one by one by iterating over them in a *for* loop. We have also coloured the markers as per the value of the *aqi*. So if the value of *aqi* is above 300, the colour is red, and so on. The signature of the *Marker* class (along with the description of the two parameters of interest, namely, *location* and *popup)* is as below:

```
class folium.map.Marker(location, popup=None, tooltip=None,
icon=None, draggable=False, **kwargs)
```

```
Parameters
•    location (tuple or list) – Latitude and Longitude of
Marker (Northing, Easting)
•    popup (string or folium.Popup, default None) – Label
for the Marker; either an escaped HTML string to initialize
folium.Popup or a folium.Popup instance.
```

The code for Step 5 is as follows:

```
###-STEP 5: Plot stations on map
centre_point = [23.25, 77.41]  # Approx over Bhopal
m2 = folium.Map(location = centre_point,
       tiles = 'Stamen Terrain',
       zoom_start= 6)
for idx, row in df1.iterrows():
    lat = row['lat']
    lon = row['lon']
    station = row['station_name'] + ' AQI=' + str(row['aqi'])
    station_aqi = row['aqi']
    if station_aqi > 300:  ## Red for very bad AQI
        pop_color = 'red'
    elif station_aqi > 200:
        pop_color = 'orange'  ## Orange for moderate AQI
    else:
        pop_color = 'green'  ## Green for good AQI
    folium.Marker(location= [lat, lon],
                  popup = station,
                  icon = folium.Icon(color = pop_color)).
add_to(m2)
m2
```